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Experiment 3

Hybrid Cloud Connectivity Using Cisco Packet Tracer

AIM:

To create a hybrid cloud network in Cisco Packet Tracer by connecting an on-premises network with a cloud network. This experiment helps in understanding how routers, switches, IP addressing, and static routing are used to establish communication between two different networks.

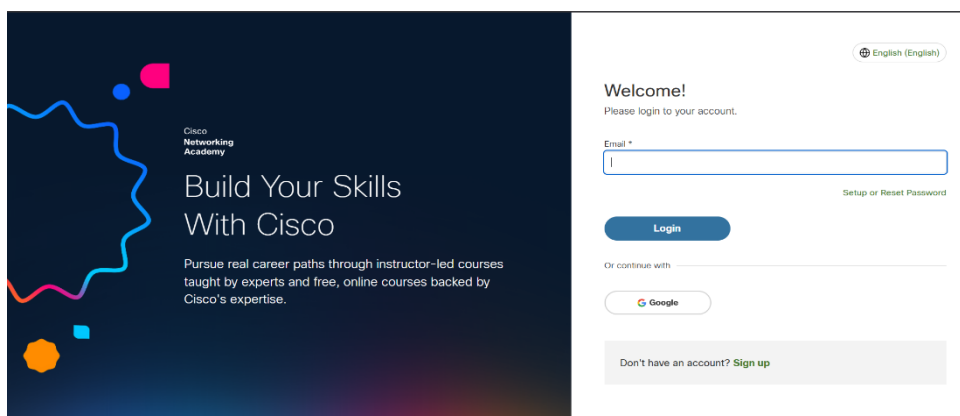
What is Hybrid Cloud?

A hybrid cloud is a combination of a private network and a public cloud. The private network belongs to an organization, while the public cloud is provided by cloud service providers such as AWS or Azure. Both networks are securely connected so that they can exchange data and resources.

PROCEDURE:

Step 1

Create a Cisco Networking Academy (NetAcad) account or sign in using your existing Google account.

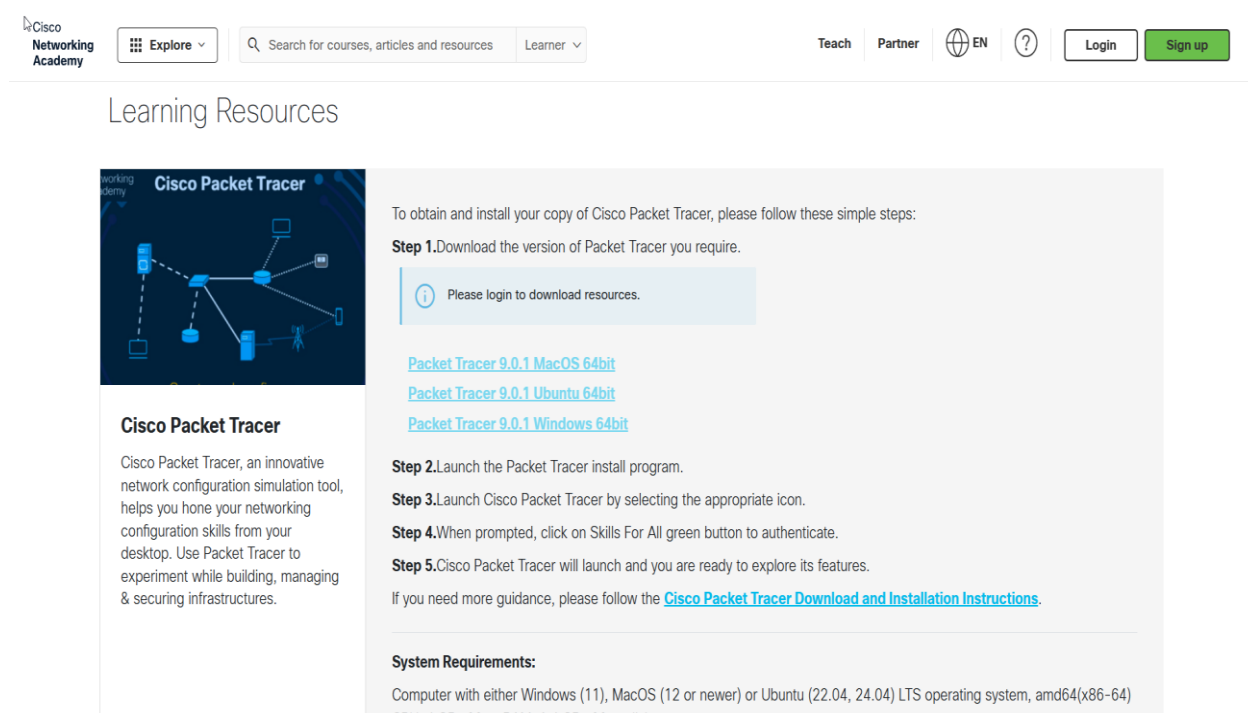


Step 2

Enroll in the free course "**Getting Started with Cisco Packet Tracer.**"

Step 3

Download the Packet Tracer software suitable for your operating system and complete the installation.



The screenshot shows the Cisco Networking Academy website. At the top, there is a navigation bar with the Cisco Networking Academy logo, an 'Explore' dropdown menu, a search bar, and a 'Learner' dropdown. On the right side of the navigation bar are links for 'Teach', 'Partner', a globe icon with 'EN', a help icon, and 'Login' and 'Sign up' buttons. Below the navigation bar is a 'Learning Resources' section. On the left, there is a card for 'Cisco Packet Tracer' with a description: 'Cisco Packet Tracer, an innovative network configuration simulation tool, helps you hone your networking configuration skills from your desktop. Use Packet Tracer to experiment while building, managing & securing infrastructures.' To the right of this card is a larger box containing instructions for downloading and installing Packet Tracer. It starts with 'To obtain and install your copy of Cisco Packet Tracer, please follow these simple steps:' followed by five steps. Step 1 is 'Download the version of Packet Tracer you require.' and includes a message 'Please login to download resources.' and three links: 'Packet Tracer 9.0.1 MacOS 64bit', 'Packet Tracer 9.0.1 Ubuntu 64bit', and 'Packet Tracer 9.0.1 Windows 64bit'. Steps 2 through 5 describe launching the install program, selecting the icon, authenticating, and launching the application. A link to 'Cisco Packet Tracer Download and Installation Instructions' is provided. Below the steps are 'System Requirements' which specify the operating system (Windows 11, MacOS 12 or newer, or Ubuntu 22.04, 24.04 LTS) and hardware (amd64/x86-64).

Cisco Packet Tracer

Cisco Packet Tracer, an innovative network configuration simulation tool, helps you hone your networking configuration skills from your desktop. Use Packet Tracer to experiment while building, managing & securing infrastructures.

To obtain and install your copy of Cisco Packet Tracer, please follow these simple steps:

Step 1.Download the version of Packet Tracer you require.

Please login to download resources.

[Packet Tracer 9.0.1 MacOS 64bit](#)
[Packet Tracer 9.0.1 Ubuntu 64bit](#)
[Packet Tracer 9.0.1 Windows 64bit](#)

Step 2.Launch the Packet Tracer install program.

Step 3.Launch Cisco Packet Tracer by selecting the appropriate icon.

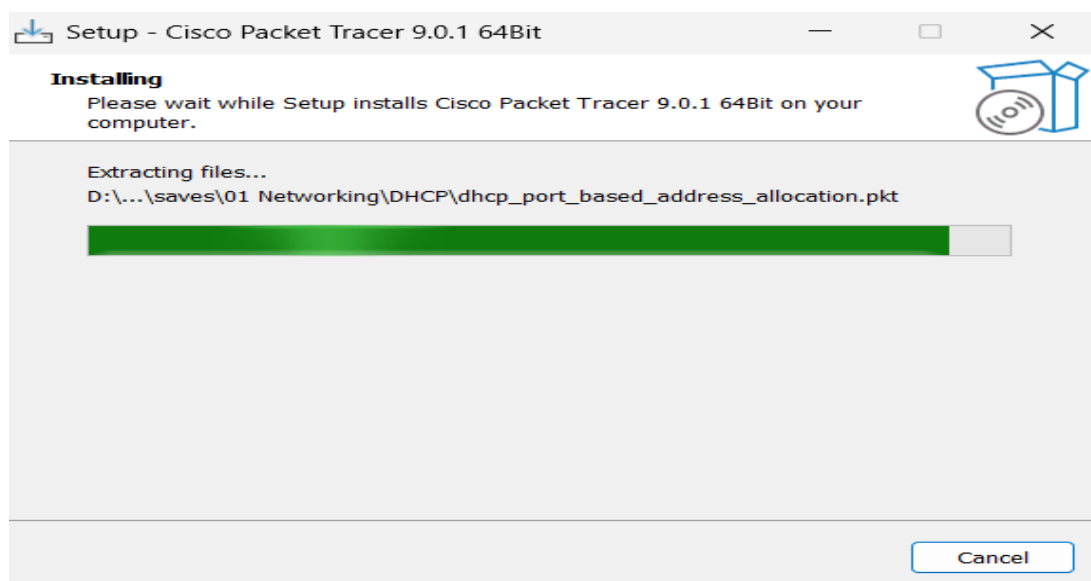
Step 4.When prompted, click on Skills For All green button to authenticate.

Step 5.Cisco Packet Tracer will launch and you are ready to explore its features.

If you need more guidance, please follow the [Cisco Packet Tracer Download and Installation Instructions](#).

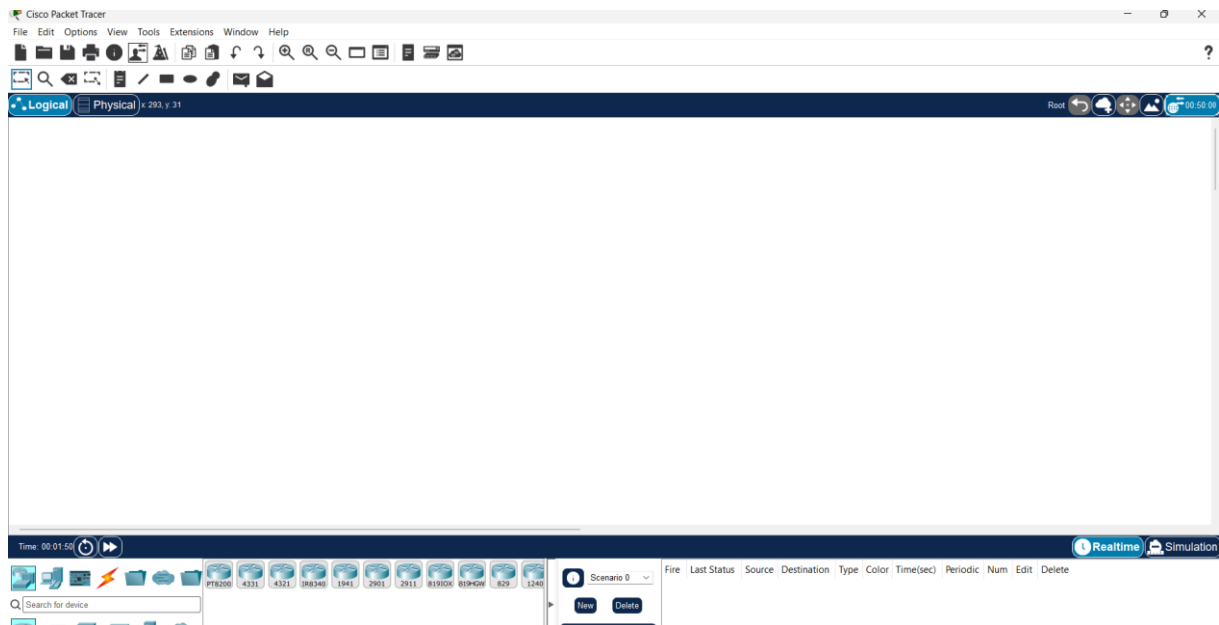
System Requirements:

Computer with either Windows (11), MacOS (12 or newer) or Ubuntu (22.04, 24.04) LTS operating system, amd64(x86-64)



Step 4

Launch Cisco Packet Tracer and log in using your NetAcad account. A blank workspace will appear



Step 5

Add the required network devices to the workspace:

- Two Routers
- Two Switches
- One PC
- One Server

Arrange the devices so that one router and switch represent the on-premises network, while the other router and switch represent the cloud network.

Step 6

Connect the devices using the appropriate cables:

- Connect the PC to Switch0.
- Connect Switch0 to Router0.

- Connect the Server to Switch1.
- Connect Switch1 to Router1.
- Finally, connect Router0 and Router1 using a crossover cable.

Step 7

Open the CLI of Router0 and configure the interfaces by assigning IP addresses to the LAN network and the router-to-router connection. Enable the interfaces using the **no shutdown** command.

Step 8

Repeat the same process for Router1 by assigning the required IP addresses and enabling both interfaces.

Step 9

Verify that all router interfaces are active using the **show ip interface brief** command. The interface status should display **up/up**.

Step 10

Configure the PC with the following details:

- IP Address: **192.168.1.10**
- Subnet Mask: **255.255.255.0**
- Default Gateway: **192.168.1.1**

Configure the Server with:

- IP Address: **10.0.0.10**
- Subnet Mask: **255.255.255.0**
- Default Gateway: **10.0.0.1**

Step 11

Configure static routes on both routers so that each router knows how to reach the other network. Save the configuration and verify that the routes are added successfully using the **show ip route** command.

Step 12

Open the Command Prompt on the PC and use the **ping** command to check communication with the server. If the ping is successful, it confirms that the hybrid cloud network is working correctly.

RESULT

The hybrid cloud network was created successfully in Cisco Packet Tracer. Communication between the on-premises PC and the cloud server was verified using the **ping** command, confirming successful network connectivity.

CONCLUSION

This experiment helped me understand how a hybrid cloud network can be simulated using Cisco Packet Tracer. I learned how to connect network devices, configure router interfaces, assign IP addresses, add static routes, and test communication between two different networks. The successful ping test confirmed that the hybrid cloud connection was established correctly.